COURSE TRACKER: SHERYIANS AI SCHOOL

Complete Statistics & ML Foundations

Compiled Topics: Z-Scores, Probability (Multiplication Rule), Bayes' Theorem, Inferential Stats, and Hypothesis Testing.

1. Z-Scores & Standardization

A **Z-Score** measures how many standard deviations a data point is from the population mean. It allows us to standardize different datasets to a standard normal distribution (Mean = 0, Standard Deviation = 1).

$$Z = (X - \mu) / \sigma$$

- **X:** Individual data point
- **μ (Mu):** Population Mean
- **σ (Sigma):** Population Standard Deviation

2. Probability & The Multiplication Rule

The multiplication rule is used to find the probability of two events occurring together, denoted as **P(A ∩ B)**.

Independent Events

Event A does **not** affect Event B.

$$P(A \cap B) = P(A) \times P(B)$$

Example: Rolling a 6 on a die and flipping a coin to get Heads.

Dependent Events

Event A **affects** the outcome of Event B.

$$P(A \cap B) = P(A) \times P(B|A)$$

Example: Drawing two cards from a deck without replacement.

Conditional Probability: The probability of B given that A has already occurred: **P(B|A) = P(A ∩ B) / P(A)**

3. Bayes' Theorem

Bayes' Theorem provides a way to revise existing predictions or theories (Prior Probability) given new or additional evidence (Likelihood).

$$P(A|B) = [\ P(B|A) \times P(A)\ ] / P(B)$$

- **P(A) [Prior]:** Initial probability of hypothesis A.
- **P(B|A) [Likelihood]:** Probability of seeing evidence B if A is true.
- **P(A|B) [Posterior]:** Updated probability of A after considering evidence B.
- **Total Probability & Unions: P(A ∪ B) = P(A) + P(B) − P(A ∩ B)**

4. Descriptive vs. Inferential Statistics

Descriptive Statistics

Summarizes and organizes characteristics of a specific dataset. Uses measures of central tendency (Mean, Median, Mode) and spread (Range, Variance, Standard Deviation).

Inferential Statistics

Uses a random sample of data taken from a population to describe and make inferences about the whole population. Relies heavily on probability theory.

5. Hypothesis Testing Framework

A systematic way to test claims or ideas about a group or population using sample data.

- **Null Hypothesis (H_0):** The default position; assumes no effect, no difference, or no relationship exists.
- **Alternative Hypothesis (H_a):** The claim you are trying to prove; assumes there is a significant effect or difference.
- **Significance Level (α):** The probability threshold (often 0.05 or 5%) at which we reject H_0.
- **Confidence Interval (1 − α):** The range of values we are confident contains the true population parameter (e.g., 95% Confidence).

Errors in Hypothesis Testing

Decision Made	Null Hypothesis (H_0) is TRUE	Null Hypothesis (H_0) is FALSE
Reject H_0	**Type I Error (α)** False Positive	**Correct Decision** True Positive (Power)
Fail to Reject H_0	**Correct Decision** True Negative	**Type II Error (β)** False Negative